Lecture 3: Adversarial Robustness

CS 580B1: Trustworthy Machine Learning

Fall 2024

Acknowledgments

Course slides derived from material created by Rajeev Alur and Osbert Bastani at UPenn (link to their course)

Slides from Adversarial Robustness - Theory and Practice tutorial at NeurIPS'18 by Zico Kolter and Aleksander Madry

Other relevant courses:

CS 329T at Stanford

CS 521 at UIUC

ECE1784H/CSC2559H at U Toronto

Course logistics

- Review due tomorrow at 11:59 PM

- Paper to review: Towards Evaluating the Robustness of Neural Networks by Nicholas Carlini and David Wagner

Quick recap

So far:

- What is trustworthy machine learning and why we care about it?

- Introduction to deep neural networks:
  - Computational model
  - Training via gradient descent and backpropagation

Agenda

- What are adversarial examples?

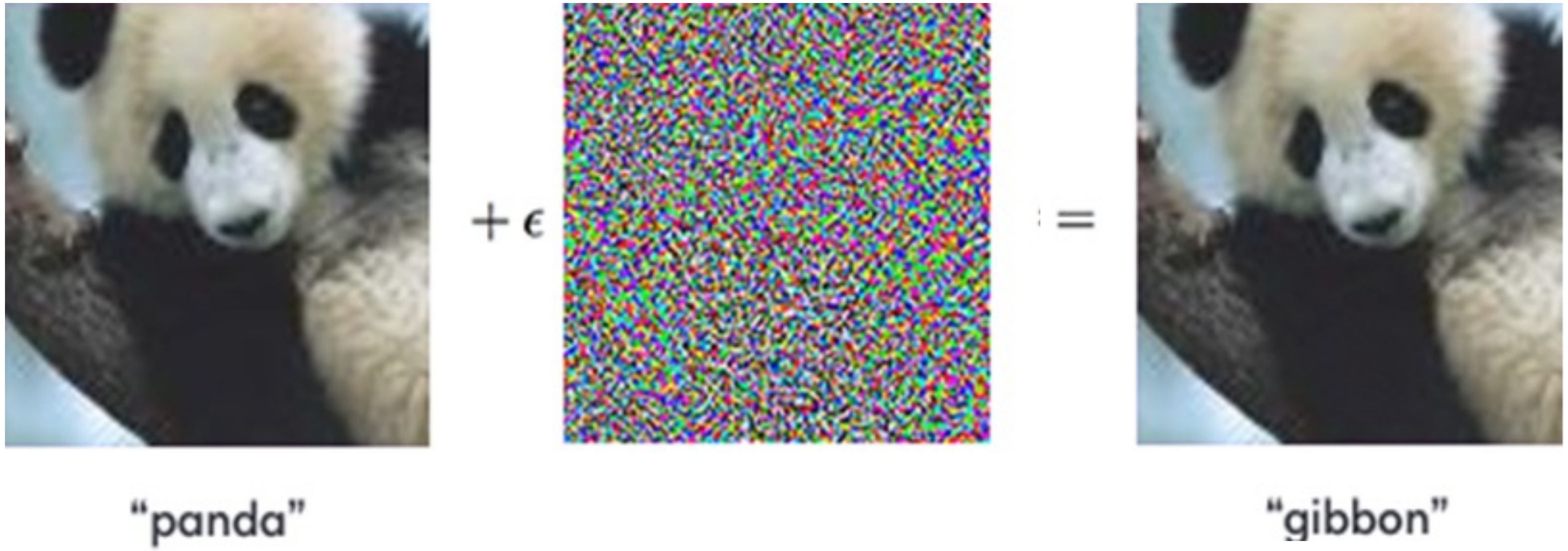
- Algorithms for finding adversarial examples

Based on papers:

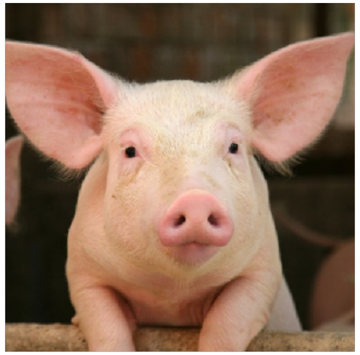
Szegedy et al., Intriguing properties of neural networks, ICLR'14

Goodfellow et al., Explaining and harnessing adversarial examples, ICLR'15

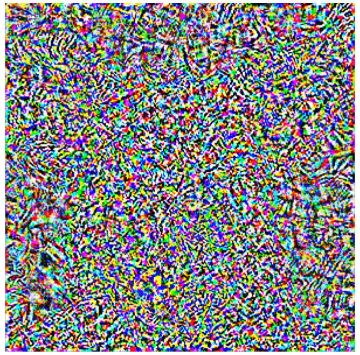
Adversarial examples



“pig” (91%) **noise (NOT random)** **“airliner” (99%)**



+ 0.005 x



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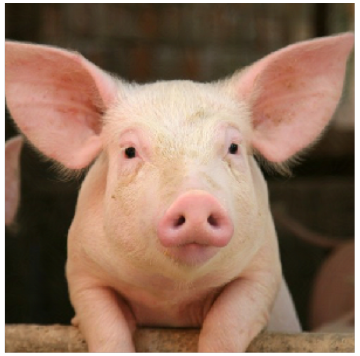


Goodfellow et al., Explaining and harnessing adversarial examples, ICLR’15
Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS’18 Tutorial

Adversarial examples: Physically realizable

Misclassified as “Speed Limit 45” sign

(a) (b) (c) (d)

Figure 4: Examples of successful impersonation and dodging attacks. Fig. (a) shows S_A (top) and S_B (bottom) dodging against DNN_B. Fig. (b)–(d) show impersonations. Impersonators carrying out the attack are shown in the top row and corresponding impersonation targets in the bottom row. Fig. (b) shows S_A impersonating Milla Jovovich (by Georges Biard; source: https://goo.gl/GlsWlC); (c) S_B impersonating S_C; and (d) S_C impersonating Carson Daly (by Anthony Quintano; source: https://goo.gl/VfnDct).

Eykholt et al., Robust Physical-World Attacks on Deep Learning Visual Classification, CVPR’18
Sharif et al., Accessorize to a Crime: Real and Stealthy Attacks on State-of-the-Art Face Recognition, CCS’16

Adversarial examples: Patch attacks

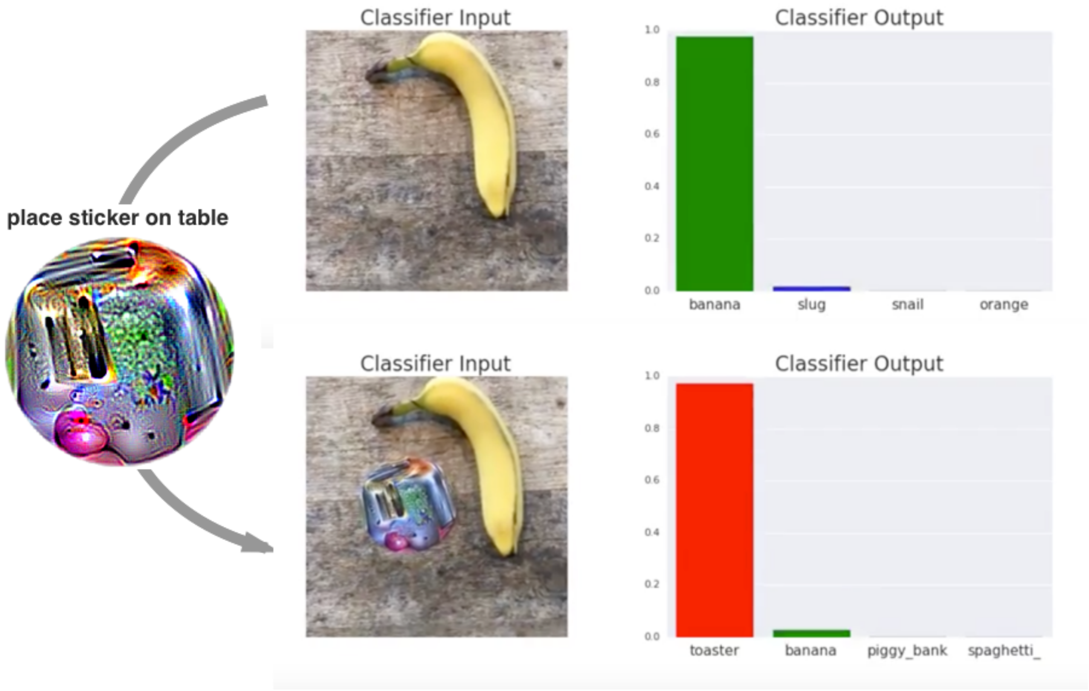
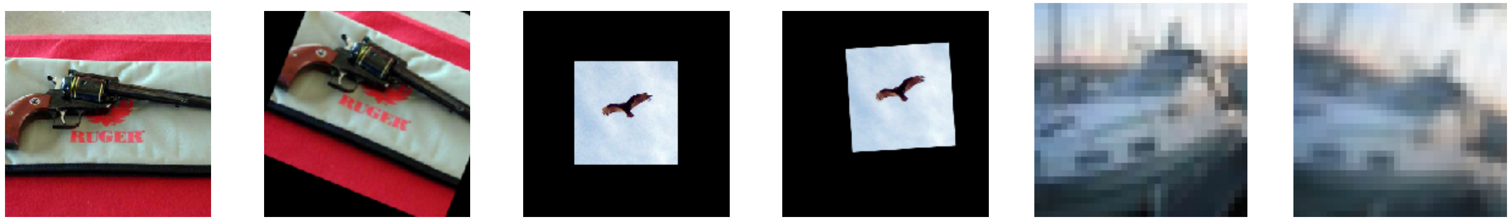


Figure 1: A real-world attack on VGG16, using a physical patch generated by the white-box ensemble method described in Section 3. When a photo of a tabletop with a banana and a notebook (top photograph) is passed through VGG16, the network reports class 'banana' with 97% confidence (top plot). If we physically place a sticker targeted to the class "toaster" on the table (bottom photograph), the photograph is classified as a toaster with 99% confidence (bottom plot). See the following video for a full demonstration: `https://youtu.be/i1sp4X57TL4`

Brown et al., Adversarial Patch, NeurIPS'17

Adversarial examples: Geometric transformations

Natural | Adversarial | Natural | Adversarial | Natural | Adversarial



“revolver” | “mousetrap” | “vulture” | “orangutan” | “ship” | “dog”

Figure 1: Examples of adversarial transformations and their predictions in the standard, "black canvas", and reflection padding setting.

Engstrom et al., Exploring the Landscape of Spatial Robustness, ICML’19

Adversarial examples: Text

South Africa's historic Soweto township marks its 100th birthday on Tuesday in a mood of optimism. 57% **World**
South Africa's historic Soweto township marks its 100th birthday on Tuesday in a moo**P** of optimism. 95% **Sci/Tech**
Chancellor Gordon Brown has sought to quell speculation over who should run the Labour Party and turned the attack on the opposition Conservatives. 75% **World**
Chancellor Gordon Brown has sought to quell speculation over who should run the Labour Party and turned the attack on the o**B**position Conservatives. 94% **Business**

Table 1: Adversarial examples with a single character change, which will be misclassified by a neural classifier.

Task: Sentiment Analysis. ***Classifier:*** CNN. ***Original label:*** 99.8% Negative. ***Adversarial label:*** 81.0% Positive.

Text: I love these ~~awful~~ **awf ul** 80's summer camp movies. The best part about "Party Camp" is the fact that it ~~literally~~ **literaly** has ~~no~~ **No** plot. The ~~cliches~~ **clichs** here are limitless: the nerds vs. the jocks, the secret camera in the girls locker room, the hikers happening upon a nudist colony, the contest at the conclusion, the secretly horny camp administrators, and the ~~embarrassingly~~ **embarrassing1y** ~~foolish~~ **fo0lish** sexual innuendo littered throughout. This movie will make you laugh, but never intentionally. I repeat, never.

Task: Sentiment Analysis. ***Classifier:*** Amazon AWS. ***Original label:*** 100% Negative. ***Adversarial label:*** 89% Positive.

Text: I watched this movie recently mainly because I am a Huge fan of Jodie Foster's. I saw this movie was made right between her 2 Oscar award winning performances, so my expectations were fairly high. ~~Unfortunately~~ **Unf0rtunately**, I thought the movie was ~~terrible~~ **terrib1e** and I'm still left wondering how she was ever persuaded to make this movie. The script is really ~~weak~~ **wea k**.

Task: Toxic Content Detection. ***Classifier:*** LSTM. ***Original label:*** 96.7% Toxic. ***Adversarial label:*** 83.5% Non-toxic.

Text: hello how are you? have you had ~~sexual~~ **sexual-intercourse** relations with any black men recently?

Task: Toxic Content Detection. ***Classifier:*** Perspective. ***Original label:*** 92% Toxic. ***Adversarial label:*** 78% Non-toxic.

Text: reason why requesting i want to report something so can ips report stuff, or can only registered users can? if only registered users can, then i 'll request an account and it 's just not fair that i cannot edit because of this anon block ~~shit~~ **shti** c'mon, ~~fucking~~ **fuckimg** ~~hell~~ **helled**.

Fig. 1. Adversarial examples against two natural language classification tasks. Replacing a fraction of the words in a document with adversarially-chosen bugs fools classifiers into predicting an incorrect label. The new document is classified correctly by humans and preserves most of the original meaning although it contains small perturbations.

Ebrahimi et al., HotFlip: White-Box Adversarial Examples for Text Classification, ACL'18
Li et al., TEXTBUGGER: Generating Adversarial Text Against Real-world Applications, NDSS'19

Adversarial examples: Speech

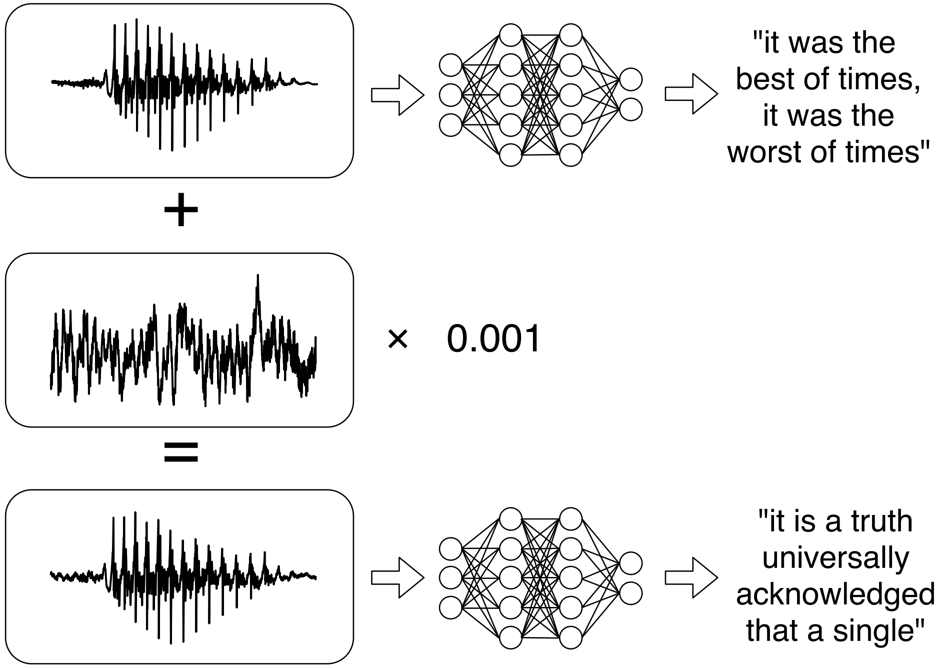


Figure 1. Illustration of our attack: given any waveform, adding a small perturbation makes the result transcribe as any desired target phrase.

Carlini and Wagner, Audio Adversarial Examples: Targeted Attacks on Speech-to-Text, DLS'18

Adversarial example: Formal definition

<u>Untargeted adv. example</u>: Given an input x and a model f, x' is an adversarial example wrt x if

$$f(x) \neq f(x')$$

and $x' \in \{x + \delta \mid \delta \in \Delta\}$

where Δ is the allowed set of perturbations

<u>Targeted adv. example</u>: Given an input x, a model f, and a target label y_{trg}, x' is a targeted adversarial example wrt x if

$$f(x) \neq f(x') = y_{trg}$$

and $x' \in \{x + \delta \mid \delta \in \Delta\}$

Perturbation sets

Δ is commonly defined using the ℓ_p norm as

$$\Delta \coloneqq \{x \in \mathbb{R}^d \mid \|x\|_p \leq \epsilon\}$$

where $\|x\|_p$ (ℓ_p norm of a real vector x) is defined as

$$\|x\|_p \coloneqq (|x_1|^p + \ldots + |x_n|^p)^{1/p}$$

Some common ℓ_p norms:

$$\|x\|_2 \coloneqq (|x_1|^2 + \ldots + |x_n|^2)^{\frac{1}{2}}$$

$$\|x\|_\infty \coloneqq \max(|x_1|, \ldots, |x_n|)$$

$$\|x\|_1 \coloneqq (|x_1| + \ldots + |x_n|)$$

$$\|x\|_0 \coloneqq (|x_1|^0 + \ldots + |x_n|^0) \text{ (using } 0^0 = 0)$$

Local robustness: Formal definition

A model f is locally robust at input x with respect to perturbation set Δ if

$$\forall \delta \in \Delta . f(x+\delta) = f(x)$$

<u>Note</u>: If model f is locally robust at x, then there exists NO adversarial example for f wrt x

Finding adversarial examples: Optimization formulation

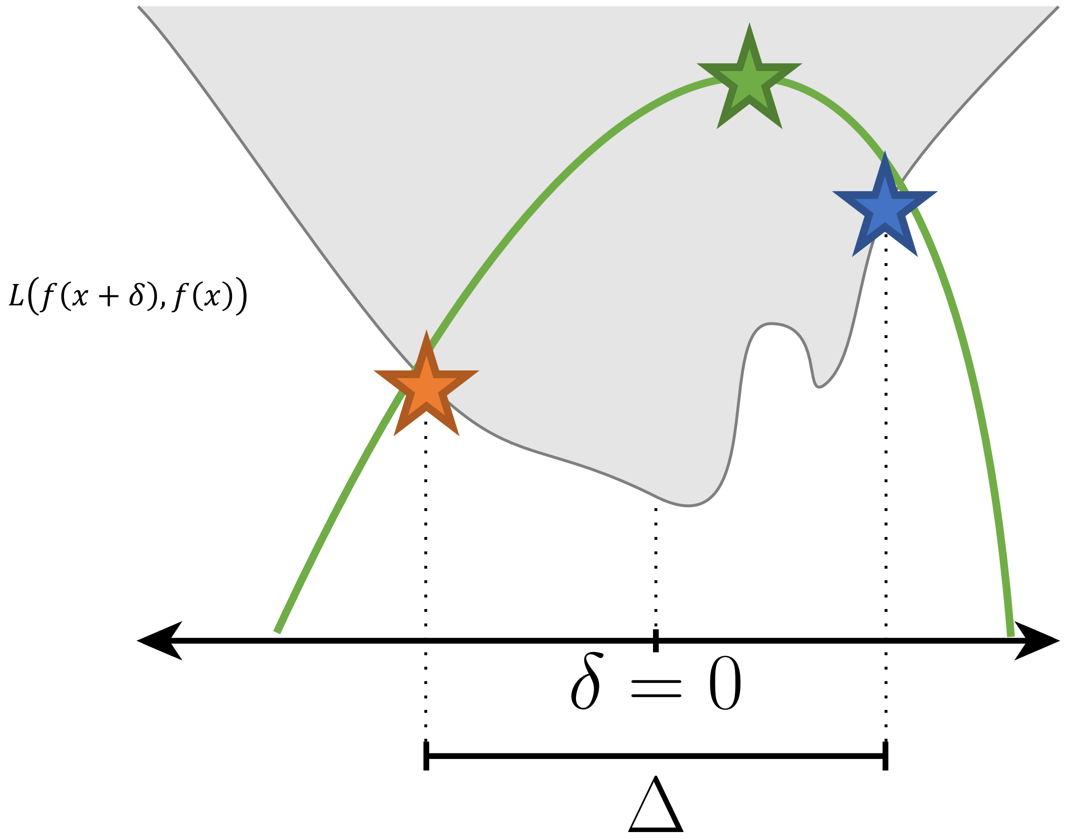
Given model f and input x, the search for an adversarial example wrt perturbation set Δ can be framed as:

$$argmax_{\delta}\ L\big(f(x+\delta), f(x)\big)$$
$$\text{such that } \delta \in \Delta$$

Note that this is a *constrained* optimization problem.

We will focus on ℓ_p norm-based perturbation sets

How to solve the optimization problem?



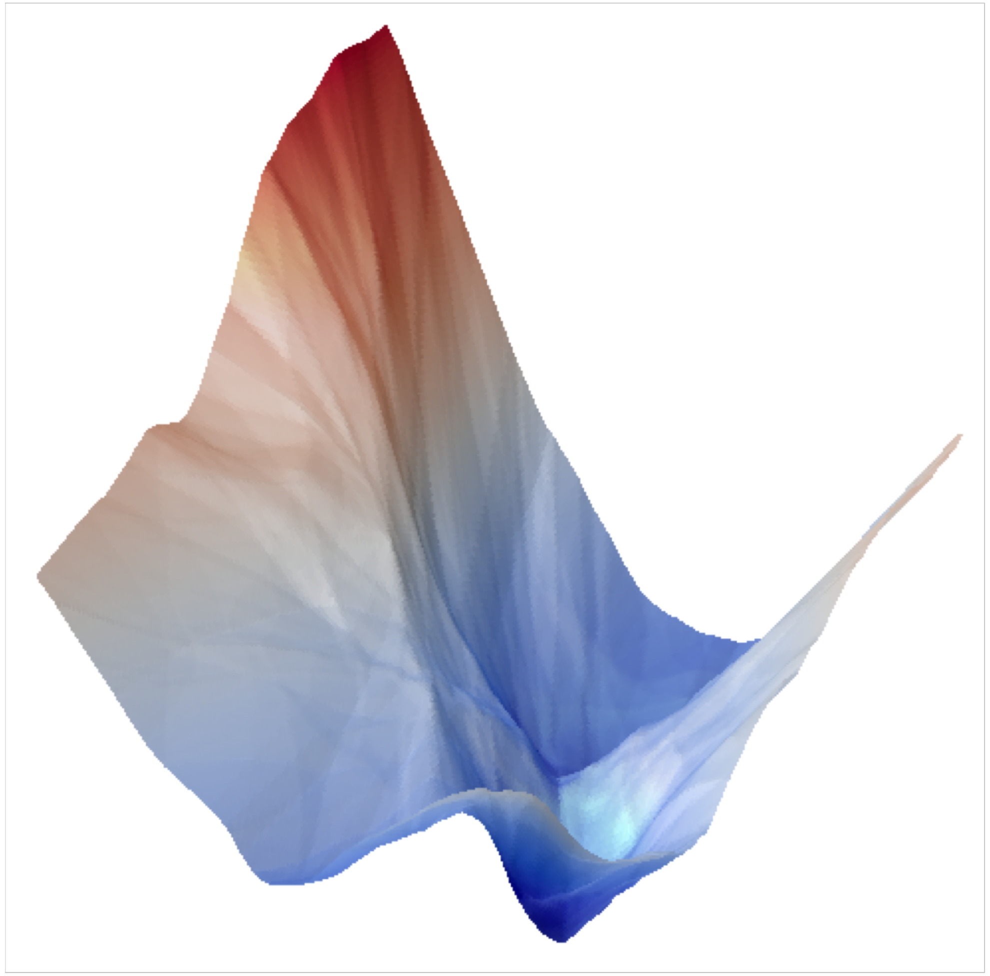
$$argmax_\delta \, L\big(f(x+\delta), f(x)\big)$$
such that $\delta \in \Delta$

1. Local search via gradients (lower bound on objective)
2. Combinatorial optimization (exact solution on objective)
3. Convex relaxation of the optimization problem (upper bound on objective)

We will focus on <u>local search via gradients.</u>

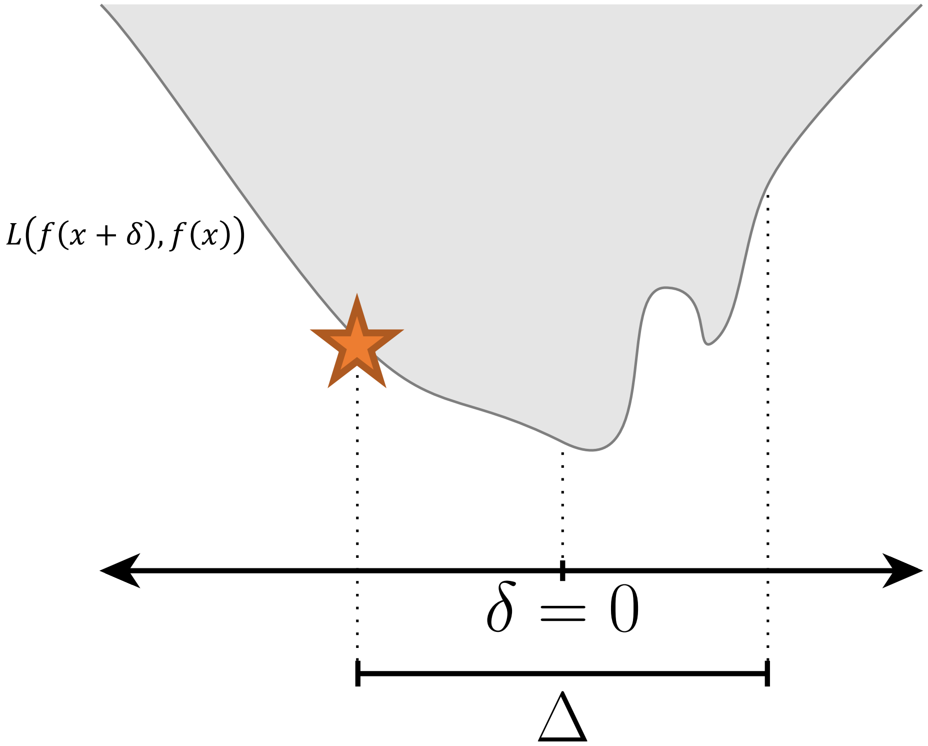
Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS'18 Tutorial

The optimization problem is hard!



Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS’18 Tutorial

Finding adversarial examples via local search



$$argmax_{\delta}\, L\big(f(x+\delta), f(x)\big)$$

such that $\delta \in \Delta$

Compute gradients via backpropagation

Cannot apply gradients repeatedly!
Recall, $\delta \in \Delta$

Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS'18 Tutorial

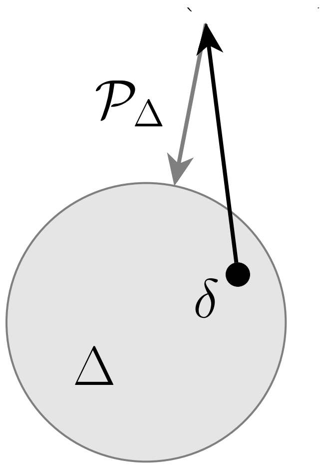
Finding adversarial examples: Projected Gradient Descent (PGD) algorithm

$\delta_0 \leftarrow Initialize()$
$t \leftarrow 0$
$Until\ convergence\ or\ budget\ exhaustion$:

$$\delta_{t+1} \leftarrow \mathcal{P}_\Delta\left(\delta_t + \eta\, \nabla_\delta\left(L\big(f(x+\delta_t), f(x)\big)\right)\right)$$

$$t \leftarrow t+1$$

$return\ (x + \delta_t)$

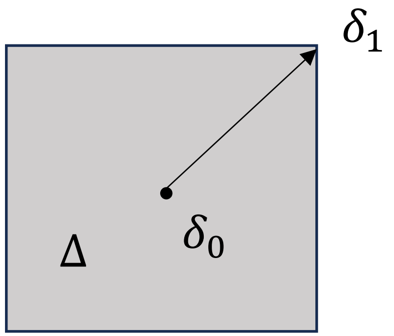


General algorithm for untargeted adversarial examples and any perturbation set Δ

Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS’18 Tutorial

Finding adversarial examples: Fast Gradient Sign Method (FGSM) algorithm

$\delta_0 \leftarrow Initialize()$
$\delta_1 \leftarrow \epsilon \cdot sign\big(\nabla_\delta L(f(x+\delta_0), f(x)\,)\big)$
$return\ (x + \delta_1)$



Always finds the "corner" of the box

- Algorithm by Goodfellow et al. for untargeted adversarial examples with perturbation set $\Delta \coloneqq \{x \in \mathbb{R}^d \mid \|x\|_\infty \leq \epsilon\}$

- Gradients wrt δ calculated using backpropagation

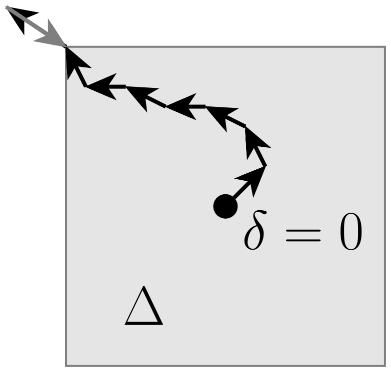
- δ_1 guaranteed to be in Δ

Goodfellow et al., Explaining and harnessing adversarial examples, ICLR'15

Why is δ_1 guaranteed to be in Δ?

Relationship between FGSM and PGD

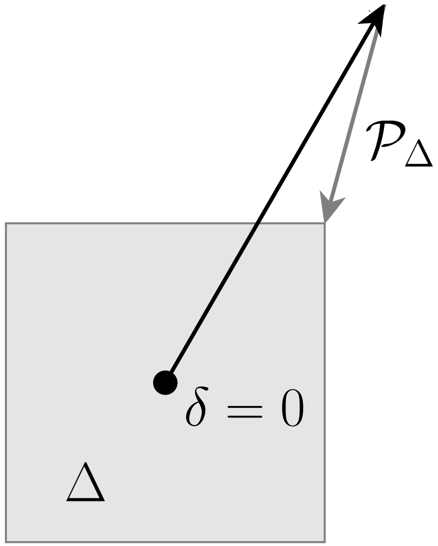
$\delta_0 \leftarrow Initialize()$
$t \leftarrow 0$
$Until\ convergence\ or\ budget\ exhaustion$:
$\qquad \delta_{t+1} \leftarrow \mathcal{P}_\Delta\left(\delta_t + \eta\, \nabla_\delta\left(L\big(f(x+\delta_t), f(x)\big)\right)\right)$
$\qquad t \leftarrow t+1$
$return\ (x+\delta_t)$



- Consider PGD where Δ is defined using ℓ_∞ norm with $\mathcal{P}_\Delta(\delta) \coloneqq Clip(\delta, [-\epsilon, \epsilon])$

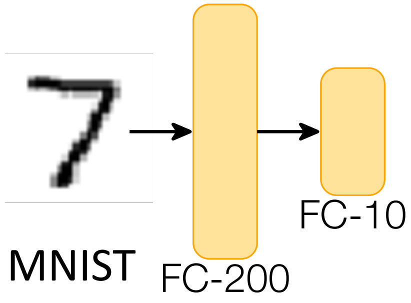
- As $\eta \to \infty$, we always reach the "corner" of the box

- PGD is slower than FGSM (requires multiple iterations), put typically able to find better optima

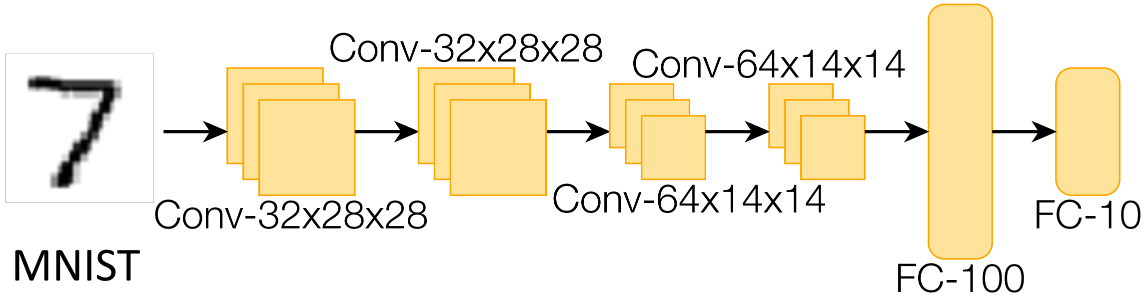


Empirical evaluation

2-layer fully connected MLP



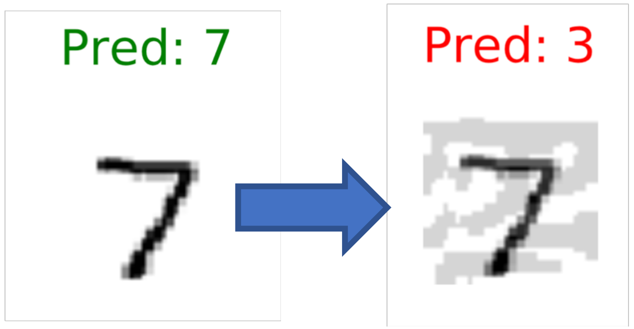
6 layer ConvNet



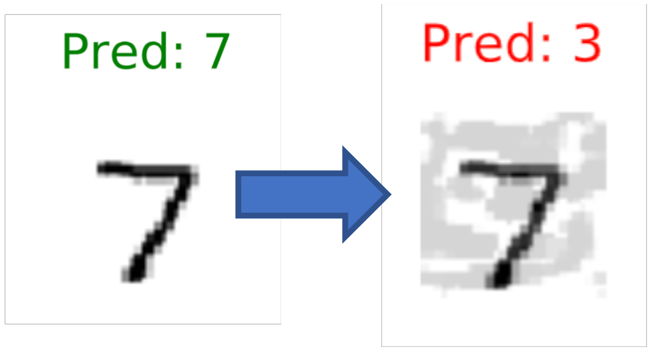
Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS’18 Tutorial

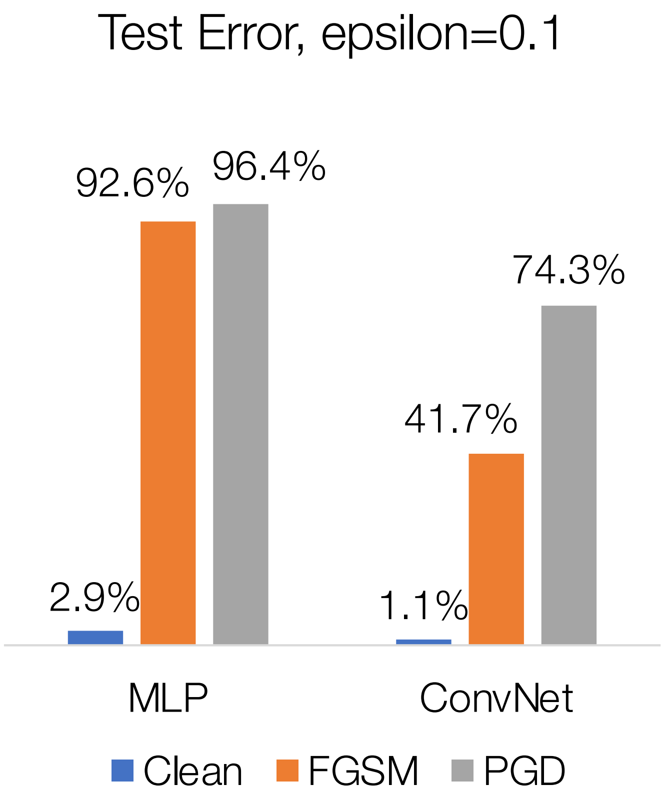
Empirical evaluation

ConvNet (FGSM):



ConvNet (PGD)





Kolter and Madry, Adversarial Robustness - Theory and Practice, NeurIPS'18 Tutorial

Finding adversarial examples: Szegedy et al.'s algorithm

$x'_0 \leftarrow Initialize()$
$t \leftarrow 0$
$Until\ convergence$:
$\quad x'_{t+1} \leftarrow x'_t - \eta\ \nabla_{x'}\big(L\big(f(x'), y_{trgt}\big)\ + c \cdot \|x'\|_p\big)$
$\quad t \leftarrow t + 1$
$return\ x'_t$

Note: Szegedy et al. actually use 2nd order optimization methods. They also constrain the inputs to lie in the unit box and perform a line search to minimize c.

Goal is to find *targeted* adversarial example wrt target y_{trgt}

Gradients calculated using backpropagation

Same adversarial example frequently "effective" on other classifiers with different architectures or trained using different datasets

Szegedy et al., Intriguing properties of neural networks, ICLR'14

Other variants

- Many variants of algorithms for finding adversarial examples
- Black-box vs white-box threat models
- Beyond ℓ_p threat models---geometric transformations, patch attacks, corruptions such as noise, blur, weather, and digital artifacts
- Discrete vs continuous inputs

Adversarial examples: Upcoming agenda

- (This week) Algorithms for finding adversarial examples
- (Next week) Defending against adversarial examples via training
- (Sept 10 and 12) Introduction to formal methods
- (Sept 17 and 19) Certifying local robustness of DNNs using formal methods